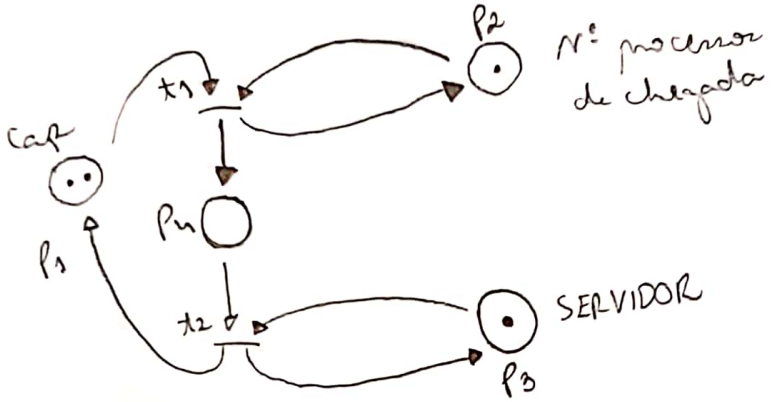
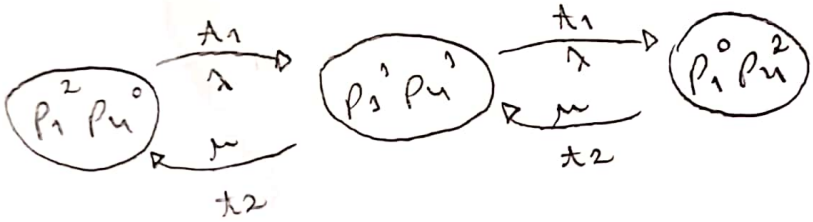
Trabalho 09 – Modelagem e Simulação

① $\lambda = 1\,min^{-1}$ $\quad \frac{1}{\mu} = 1\,min \rightarrow \mu = 1\,min^{-1}$

a)



b)



c)

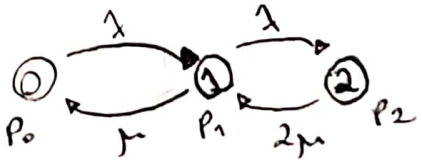


d)

② M/M/2/2

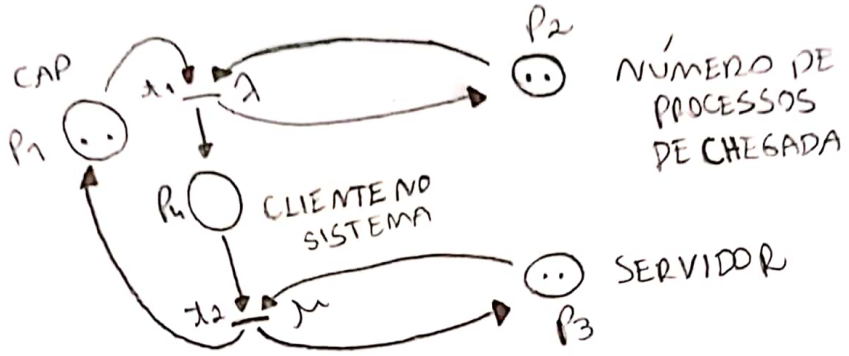
$\lambda = 1\,min^{-1}$

$\frac{1}{\mu} = 1\,min \rightarrow \mu = 1\,min^{-1}$



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a)



b)



c)



d) Considerando $\lambda = \mu = 1\ min^{-1}$

Est. 0: $\mu \cdot P_1 = \lambda P_0$

Est. 1: $\lambda P_0 + 2\mu \cdot P_2 = \mu P_1 + \lambda \cdot P_1$

Est 2: $\lambda \cdot P_1 = 2\mu \cdot P_2$

Normalização: $P_0 + P_1 + P_2 = 1$

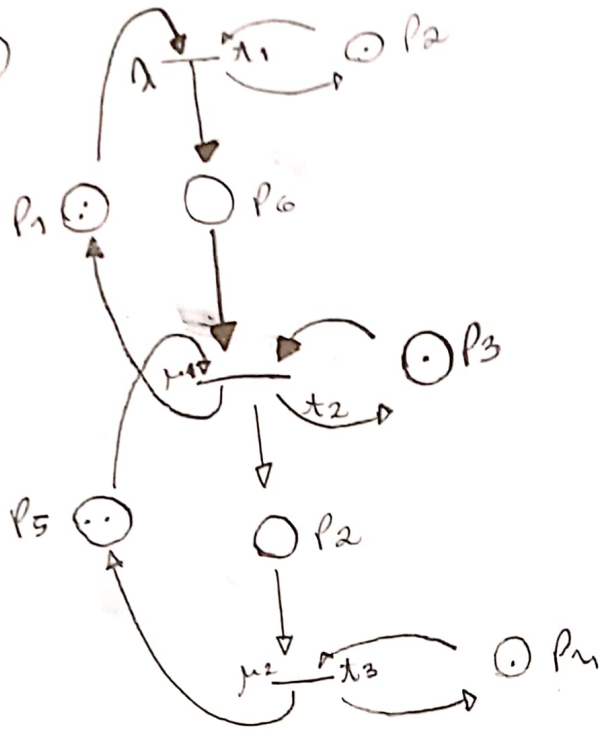
$P_0 = \frac{2}{5} \qquad P_1 = \frac{2}{5} \qquad P_2 = \frac{1}{5}$

Porcentagem de clientes atendidos será $P_0 + P_1 = 80\%$.

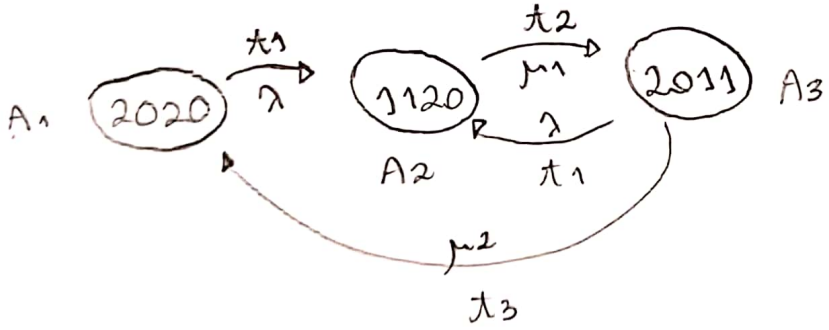
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Continuação Trabalho 9 - Modelagem e Simulação

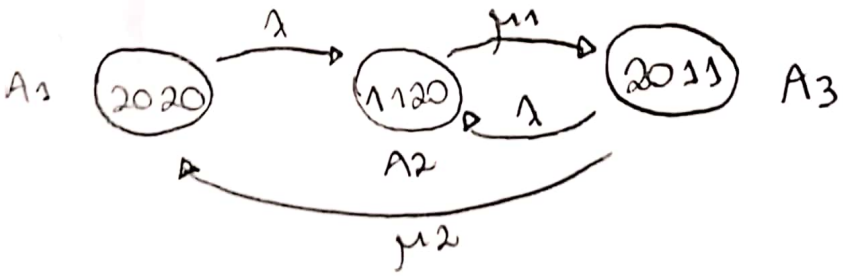
3)



a) Os estados são representados pela quantidade de fichas em P_1, P_6, P_5 e P_7, respectivamente.



b)



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